Statement of Work

Cooling Tower, Roofing, and Structural Construction

SECTION I Background

1.1 The Department of Treasury's Bureau of Engraving and Printing (BEP), Washington DC Facility (DCF) is comprised of the Main Building and the Annex Building. The mission of the BEP is to manufacture and produce United States currency.

The Annex Building was erected in 1937 and features a limestone façade, constructed entirely of reinforced concrete. The Annex Building features a central corridor with five wings that span from north to south along this line. The building is between 13th and 14th Streets and C and D Streets in the southwest quadrant of Washington, DC.

The Main Building, constructed in 1914, features a neoclassical design with an exterior of Indiana Limestone and granite. It is located between 14th and 15th Streets in the Southwest quadrant of Washington, DC. This project will take place at the Main Building in the Plant location

1.2 This Statement of Work (SOW) defines the scope, this project is a the design build project, requirements, and deliverables, including, but not limited to, a complete assessment of the scope of work, recommendation of the scope of work, demolition, and the replacement of two existing Evapco cooling towers (Model: AT 424-936) **(BRAND NAME OR EQUAL)**, and replacement of one BAC cooling tower (model number is VXT-800CR from Baltimore air coil company, Serial number is 84-5979) **(BRAND NAME OR EQUAL)** associated structural steel platform, roofing replacement, associated structural components, associated mechanical components, associated electrical components, pump, motors, piping, chemical feed, fire/smoke detector system, condenser water motor/pumps, electrical circuit breaker, piping insulation, SCADA integration, and associated control systems. Contractor should ensure that the project adheres to all guidelines and code, including but not limited to, the International Plumbing Code (IPC), the American Society of Heating, Refrigeration, and Air Conditioning Engineers (ASHRAE) standard and guideline, the American Society of Mechanical Engineers (ASME) codes and standards and the National Electrical Code, etc.

The contractor shall provide As-built drawings and updated schematics for this project. The contractor shall provide an integration and SCADA commissioning, demolition, and disposal report. The contractor should provide equipment, structural, and all necessary submittals and cut sheets for all replacement components. The work shall be performed by a qualified HVAC/mechanical licensed contractor with demonstrated experience in large-scale condenser water system installations, including the integration of cooling towers with SCADA systems. The As-built drawing needs to be accepted before the start of construction.

Based on the data published by the American Society of Heating, Refrigeration, and Air Conditioning Engineers (ASHRAE), the existing condenser pump (4), Cooling Towers (2), condenser piping, insulation, and controls have exceeded their life expectancy. Also, the existing structural steel platform that based on available records seems to be built in the mid 60's, and gone through major restoration activities in the mid 90's appears to be in a corroded and rusted conditions due to the age, weather, and existing cooling towers leaking conditions, and needs to be replaced with a new structural steel platform.

SECTION II Purpose

2.1 **Product, Service or Outcome Needed.**

The objective of this agreement is to obtain construction services for the removal, disposal, and replacement of the cooling tower, associated structural steel platform, mechanical, pump, motors, piping, chemical feed, fire, and control systems located in the BEP main building.

2.2 **Scope of the Product, Service, or Outcome.**

The contractor shall provide all personnel, equipment, supplies, facilities, transportation, tools, materials, supervision, and other items and non-personal services necessary to perform a survey and provide approved drawings for the existing structural steel platform and then the demolition for this structural steel platform, mechanical, electrical, integration, and HVAC Replacements as defined in this Statement of Work, except for those items specified as government-furnished property and services. The contractor shall perform to the standards and provide service in accordance with the SOW. The contractor should not disrupt the agency's mission during the project's duration.

Non-Personal Service statement: The Contractor employees performing services under this Contract will be controlled, directed, and supervised at all times by management personnel of the Contractor. Contractor management will ensure that employees properly comply with the performance work standards outlined in the Contract. Contractor employees will perform their duties independently of, and without the supervision of, any Government official or other Government Contractor. The tasks, duties, and responsibilities outlined in this Contract may not be interpreted or implemented in any manner that results in any Contractor employee creating or modifying Federal policy, obligating the appropriated funds of the United States Government, overseeing the work of Federal employees, providing direct personal services to any Federal employee, or otherwise violating the prohibitions outlined in Parts 7.5 and 37.1 of the FAR.

In accordance with FAR 36.204, Disclosure of the Magnitude of Construction Projects, advance notice and solicitations shall state the magnitude of the requirement in terms of physical characteristics and estimated price range. <u>The estimated price range is between $2,000,000 and $4,000,000.</u>

SECTION III Technical Requirements/Tasks/Outcomes

3.1 **<u>Specific Tasks</u>**

3.2 **<u>Basic Services.</u>**

The contractor shall provide services for all phases of the project, and each phase shall be subject to BEP approval at designated project milestones. The BEP shall remain fully operational during all phases of construction. This project includes (but is not limited to) all labor, materials, supplies, equipment, services, training, communications, transportations, shipping, supporting steel structure, HVAC controls, operation and maintenance manuals, as-built drawings, construction, testing, adjusting, balancing, and commissioning. In conjunction with the aforementioned, the Contractor shall perform the following as further detailed in the Statement of Work (SOW). Plans are not available for areas in which work is to be performed, but the BEP will try to provide, but cannot guarantee, archive drawings for “Informational Purposes Only”. It is the Contractor’s responsibility to gather all information necessary for the successful completion of this project, including but not

limited to a thorough understanding of the existing conditions under which the work will be performed. Before commencing the work, the Contractor shall conduct the following

3.2.1.1 Visit and thoroughly survey the project site and all existing conditions.

3.2.1.2 Become familiar with the project layout, conditions, equipment, components, and other attributes having or likely to have an impact on the project.

3.2.1.3 Verify existing capacities, make recommendations and provide submittals of proposed units inclusive of proposed capacities.

3.2.1.4 Contractor shall be responsible for Lead Based Paint (LBP) identification, testing/analysis and all associated LBP abatement procedures and requirements. If LBP is located, the Contractor shall note the LBP location on the project drawings, draft a report detailing the location(s) of the LBP and draft a detailed LBP Abatement Plan. The LBP Abatement plan shall be forwarded to the Office of Environment, Health & Safety (OEHS) for review and approval at least two (2) weeks prior to beginning any abatement procedures. No LBP abatements will proceed w/o written approval from the OEHS. The removal of the waste LBP shall be performed by the Contractor IAW EPA and DC Hazardous Waste Disposal regulations and per Section 0113000 - Removal and Disposal of Lead-Based Paint.

3.3 **Survey / Demolition Work**

1. **Mechanical/ Electrical/ Controls**-
   - Demolish two existing Evapco cooling towers (Model: AT 424-936), one Baltimore Air Coil cooling tower (model number is VXT-800CR from BAC, Serial number is 84-5979), including but not limited to associated mechanical, support structural, controls, and electrical components.
   - Demolish existing condenser water pumps (Pumps; 75 HP, 230/460V, 1780 RPM) and (Pump 100 HP, 230/460V, 1780), but not limited to any associated existing piping and electrical infrastructure, etc.
   - Demolish the existing condenser water motor/pump (1B, 2B, 3B, 4B), including but not limited to electrical components, piping, motors, insulation, VFDs, controls, etc.
   - Demolish existing condenser water piping from pumps (1B, 2B, 3B, 4B) to the cooling towers
   - Demolish the existing condenser chemical feed system and all associated components.
   - Demolish and remove the abandoned cooling tower labeled as Tower #5. Demo all associated items, including but not limited to piping, electrical components, and support structures, shall also be removed, and the area restored to a safe and code-compliant condition
   - De-integrate and demolish all outdated controls associated with the existing cooling towers, valves, pumps, etc.
   - Demolish any electrical components associated with this project, including but not limited to circuit breakers, wiring, capacitors, relays, fuses, electrical panel, control valve, sensors, low voltage components, etc.

- Demolish all existing controls that are associated with the condenser water pump, cooling tower, piping, fire alarm system, Scada system, VFD, raceway, including but not limited to valves, actuators, control panels, switches, sensors, fire alarm smoke system, relays, contactors, limit switches, leak detectors, etc.

- Demolish the existing plate and frame heat exchanger and the existing economizer pumps and associated controls.

- Demolish the existing roofing and all associated with it.

2. Structural/ Civil-

- Survey existing structural steel platform and prepare approved set of drawings to all the existing steel structural members such as, but not limited to, girders, beams, braces, column stub ups, connections, grating, ladders, stairs, handrails and guardrails … etc.

- Prepare and submit a signed and sealed drawings and calculations to include all the survey outcome indicated in 3.3.8 along with all related layout, dimensions, plans, elevations, structural details to BEP for review. If existing structural members are different from steel sections provided on the latest AISC Manual, then new drawings shall include equivalent and updated steel structure sections that have equivalent or larger capacities relative to the existing structure.

- Demolish, remove, salvage and dispose existing structural steel platform as described in 3.3.8 in accordance with all the governed environmental and local regulations and standards.

3.4 **New Work**

1. Mechanical/ Electrical-

- The capacities of all newly installed mechanical, electrical, controls, fire, and piping systems shall, at a minimum, match those of existing demolished units. Prior to commencing work, the contractor shall verify the existing requirements and submit submittals of proposed units, including proposed capacities.

- Newly installed interior/exterior piping shall match existing demolished piping dimensions.

- The Contractor shall provide temporary AC Units during the duration of this project before the start of work, as needed to satisfy the building current load.

- The Contractor shall select equipment that meets Energy Star/FEMP requirements as mandated by Federal Laws (where applicable). BEP requires equipment to adhere to new guidelines.

- The Contractor shall retain the services of a licensed electrical contractor to route wiring and other electrical connections for demolition and installation. The Electrical Contractor shall reuse existing wiring to the maximum extent possible and modify electrical services where necessary. All work shall adhere to the equipment manufacturers' wiring diagrams and the National Electric Code (NEC).

- The Electrical Contractor shall furnish and install electrical supply, panels, breakers, etc., where required.
- The Contractor is responsible for any required permitting for cranes, hoists, or lifts, etc.
- Conduct a comprehensive assessment of the existing mechanical, electrical, and control systems before commencing work.
- Provide a detailed recommendation report, including specifications for replacement equipment and systems.
- Obtain all necessary permits and coordinate inspections with local authorities having jurisdiction.
- Ensure all new systems comply with ASHRAE, NFPA, NEC, and relevant local building codes.
- Coordinate all shutdowns and equipment transitions to avoid disruption to building operations.
- Provide complete documentation, testing, commissioning, and training for all installed systems.
- Furnish and install two new cooling towers of the same model or approved equivalent (Model: AT 424-936) meeting all manufacturer specifications, structural support and site performance requirements
- Furnish and Install new existing plate and frame heat exchanger and the existing economizer pumps and associated controls
- Furnish and install one Baltimore Air Coil cooling tower (model number is VXT-800CR from BAC, Serial number is 84-5979) meeting all manufacturer specifications, structural support and site performance requirements
- Furnish and install two new upgraded pumps of the same size, power rating, and compatibility with the existing system.
- Ensure all new piping is restored to up-to-date insulation. The IMC specifies that pipe insulation must conform to IECC, be tested by ASTM E84 or UL 723, and have a maximum flame spread index of 25 and smoke-developed index.
- Furnish and install four (4) new upgraded condenser water pumps (Pumps 1B & 2B; 75 HP, 230/460V, 1780 RPM) and (Pump 3B & 4B; 100 HP, 230/460V, 1780 RPM) or equivalent, fully compatible with existing piping and electrical infrastructure. Ensure Pump 1B & 2B are cross-connected so that it can operate on either dedicated chiller. Ensure that Pumps 3B and 4B are cross-connected, so that it can operate on either dedicated chiller.
- Furnish and install new compatible condenser water piping from pumps (1B, 2B, 3B, 4B) to the cooling towers and install new piping that is compatible in size, material, and function

with the previous configuration. Ensure this new piping is a manifold that allows either chiller to operate when in operating/run mode and is not dedicated to one chiller. This all-new piping shall comply with applicable mechanical codes and site design standards.

- Furnish and install new piping that is compatible in size, material, and function with the previous configuration. All new piping shall comply with applicable mechanical codes and site design standards.

- Furnish and install a new chemical feed system compatible with current water treatment requirements. The system shall include all necessary tanks, dosing pumps, sensors, and controllers.

- Furnish and install a new compatible control system that integrates with the facility's SCADA system. All control components and software must be installed and tested by a certified SCADA technician. All fire protection sensors, alarms, or connected systems impacted by this work shall also be reviewed and integrated as needed.

- Furnish and install a new plate and frame heat exchanger, and the existing economizer pumps and associated controls should be replaced

2. HVAC Controls

- The government will provide the contractor with the Controls and Control Panel Specifications guideline for their use and reference

- Furnish and install all associated mechanical controls necessary for properly operating, controlling, alarming, and monitoring the HVAC systems and their related components. Control items to address shall include but are not limited to the following: VFD, smoke detectors, HVAC fan motor startup/ shutdown, and interface with existing fire detection system and fire alarm panel, condenser pumps, etc

- Furnish and install a new compatible control system that integrates with the facility's SCADA system. All control components and software must be installed and tested by a certified SCADA technician. All fire protection sensors, alarms, or connected systems impacted by this work shall also be reviewed and integrated as needed.

- All devices shall utilize SCADA communication protocol and shall have manufacturer produced Protocol Implementation Conformity Statement (PICS) submitted and approved prior to equipment approval or release for purchase.

- A building automation system schedule shall be created, and integrated to SCADA, for each necessary mechanical/ hvac equipment. The schedule shall use existing Holiday, 24/7 Operation Override, and Year End Shutdown (YES) Calendars.

- Integrate HVAC controls into the existing Supervisory Control and Data Acquisition (SCADA) system.

- Integration of HVAC controls shall be coordinated with the SCADA System Manager and shall be consistent with other existing HVAC units.

- The equipment shall have BACnet TCP/IP based communication or contractor may utilize a supervisor/gateway/or BACnet router to bridge BACnet RS-485 network to the BEP IP network. The Contractor shall be responsible for properly terminating and shielding the RS-485 communication trunk.

- Contractor to provide, label, terminate, test and certify any needed IT Network drops back to nearest enterprise network closet form the controller(s).

- The existing SCADA is a group of servers and clients with primary, redundant, and Historian SQL servers, and multiple SCADA clients. All reside on the BEP isolated network. Below is the following configuration of the SCADA system

- The SCADA system is running CITECT 8.X (2016) with redundancy failover.

- All the graphics, menu tree structure, alarm points, and trend data reside at the server level and shall be updated for new equipment.

- Historical trend data is stored on the Primary and Redundant SCADA Trend servers and is transferred to the Historian SQL server, where reports are generated.

- The existing HVAC controls utilized in the building are Facility Explorer.

- Field and panel wiring shall match the existing color coding used in existing HVAC control panels. All components and wiring shall be submitted for approval.

- Contractor shall provide one spare controller of each type, as well and one spare room sensor.

- Contractor must use controllers compatible with software existing at BEP. Otherwise, contractor must provide any and all software, licenses, cables to program/commission the controller.

- Contractors shall use BEP laptops to do initial programming, checkout, and commissioning of the controllers at BEP.

- No Contractor laptops are allowed to connect to any electronic devices at BEP. Including new HVAC controls. (IT security requirement).

- Install a new compatible control system that integrates with the facility's SCADA system. All control components and software must be installed and tested by a certified SCADA technician. All fire protection sensors, alarms, or connected systems impacted by this work shall also be reviewed and integrated as needed.

3. New structural steel platform

- The Contractor shall submit approved shop drawings for the new structural steel platform for the new cooling towers, as per the submitted and reviewed set of drawings listed in Section 3.3.9.

- The Contractor shall erect and install a new structural steel platform for the new cooling towers, as per the approved shop drawings and the submitted and reviewed set of drawings listed in Section 3.3.9.

- Contractor shall submit all necessary testing reports from an independent third-party certified testing agency for the newly constructed structural steel platform to BEP for review.

- Contractor shall provide all the equipment and scaffolding systems with all means and methods required for this project; and shall be responsible and liable to comply with all safety codes, standards, and regulations, and protect pedestrians, surrounding properties, and trees. The Contractor shall also be responsible for securing the appropriate permits from Washington, D.C., DC DOT, and local AHJ as needed.

- Coordinate and confirm the exact locations of the new cooling towers on the new platform and provide all necessary support to secure this equipment.

a. <u>Quality Requirements</u>

Testing and Inspection - The Contractor is responsible for testing and inspections. The Contractor will submit names of all testing services to the COR for approval prior to any testing. All inspections and tests must be performed by an independent, third part firm/company. The independent, third-party firm/company shall be appropriately registered in the District of Columbia. The Contractor shall engage only in inspections and testing services companies, including independent testing laboratories that are prequalified as complying with the American Council of Independent Laboratories "Recommended Requirements for Independent Laboratory Qualification" and that specialize in the types of inspections and tests to be performed.

b. The Contractor shall submit a Schedule of Tests and Inspections. The Schedule of Tests and Inspections shall be prepared in tabular form and include the following

I. Specification section Number and Title
II. Description of test and inspection
III. Identification of applicable standards
IV. Identification of test and inspection methods
V. Number of tests and inspections required.
VI. Entity responsible for performing tests and inspections.

c. All tests and inspections shall be documented by certified written reports and test results. The independent, third part individual firm and/or testing company shall distribute copies of all testing and inspection reports to the COR, and the Contractor. The Contractor shall make additional copies as required and distribute to all affected Sub-Contractors and suppliers.

d. <u>Adherence to the Scope of Work:</u> Although this SOW identifies specific elements of construction, it is the Contractor's responsibility to provide all elements which are incidental for the functioning of the work to be provided. It is not intended for other deficiencies not related to the work identified in this scope, to be corrected. The Contractor shall limit its efforts only to work identified in this SOW.

e. <u>Storing Materials:</u> The Contractor shall be responsible for receiving, storing, and securing of all materials, equipment, and other items to be used in accomplishing the

work. The Contractor shall be responsible for all equipment and materials brought onto the BEP property and shall replace damaged and/or stolen items at its own expense.

f. **Incidental Cutting and Patching:** The Contractor shall be responsible for all cutting and patching incidental to the work described herein. The Contractor shall be responsible for identifying the location of existing utilities and for avoiding damage while completing this work. The Contractor shall promptly repair and/or replace any damage to structures, elements, utilities, etc., occurring due to construction, the cost which shall be borne by the Contractor.

g. **Coordination of Work Existing Utility Interruptions:** Existing Utility Interruptions The Contractor shall not interrupt any utilities serving the facilities occupied by the BEP unless permitted under the following conditions and then only after arranging to provide temporary utility services to maintain facility function.

- Notify the BEP COR, not less than **seven (7) calendar days** in advance of the proposed utility interruption.
- Do not proceed with utility interruptions without notification to the COR and/or the CO

h. **Pre-Construction Meeting:** A pre-construction meeting is required. The meeting will include coordination with the BEP, project procedures, staging, phasing, sequencing, lines of communications, project deliverables, roles and responsibilities, safety and quality requirements.

i. **Site Visits:** Site visits and construction observations are required once construction commences, in coordination with the BEP. The Contractor shall prepare an agenda for the weekly progress meetings, compile the meeting minutes and distribute to the team members and attendees within three (3) calendar days after the meeting. These weekly progress meetings shall include representatives from the Contractor and BEP team.

j. **Parking:** There is no on-site construction parking, but vehicles are permitted to drop-off materials upon request.

k. **Dust Control:** Dust control measures shall be implemented to minimize dust from traveling within the designated area.

l. Occupant Protection must be implemented within the BEP buildings by installing safety cones, rails or other barriers to prevent injury from unauthorized entry of personnel into the work area.

m. Upon completion of work, the Contractor shall provide training to the BEP Power Plant and Facilities Maintenance personnel.

n. Prior to commencing demolition or construction activities, the Contractor shall submit for review and approval for the following

I. Preliminary Working Schedule-Due 2 weeks after the kickoff meeting.
II. Safety & Fire Protection Plan
III. Activity Hazard Analyses (AHAs)
IV. Crane Lift Plan

V. Fall Protection Plan
VI. Quality Control Plan
VII. Phasing/Staging Plan
VIII. Installers' Qualifications
IX. Hazardous Waste Plan (if applicable)

o. Construction Documents Criteria

i. Action Submittals The Contractor shall provide product data submittals for review and approval by the BEP before the purchase and installation of all new equipment.

ii. As-Built Drawings are to be submitted fourteen (14) calendar days before requesting Substantial Completion walk-through. As-built drawings shall be in PDF and AutoCad 2021.

iii. Construction Activities Arrange deliveries of products in accordance with construction schedules. Coordinate to avoid conflict with work and conditions at the site. Contractor shall minimize disruption to the daily operations of the BEP as much as possible.

iv. Deliver products and materials in an undamaged condition, in the manufacturer's original container and/or packaging, with identifying labels intact and legible. Immediately on delivery inspect shipments to assure compliance with requirements of contract documents and approved submittals, and that products are properly protected and undamaged.

v. Construction deliveries will be permitted 0600 am to 1400 pm, Monday through Friday, provided parking and other BEP activities are not disrupted. When the COR determines the deliveries will be disruptive, deliveries must be scheduled during the weekend hours. All deliveries must be scheduled with the COR and/or CC within a minimum of 48 hours in advance of the delivery. Freight elevator is available to use.

vi. Construction Work Schedule The Contractor shall submit a construction schedule to the CO and/or COR for approval. The schedule shall account for all divisions of work identified in the specifications and drawings. The schedule shall clearly show the completion date, duration and any milestones outlined in the specifications. The Contractor shall also ensure that the schedule is updated regularly or as specified by the COR.

vii. Request for Outages At least three (3) weeks prior to the delivery of a large piece of equipment, and/or commencement of activities that will impact BEP's daily operations, such as utility outages, blocking or obstructing access through corridors, etc., the Contractor shall submit in writing to the CO and/or COR his/her schedule for this work. If this work will cause an interruption of necessary services, the work shall be conducted during off-hours, weekends, or holidays.

viii. General The Contractor shall submit all Request for Information (RFI) in writing to the CO and/or COR and allow at least seven (7) calendar days for the COR to respond.

4. Supervision of Contractor's Employees Personnel assigned to render services under this contract shall always be employees of the Contractor and under the direction and control of the Contractor. Notwithstanding any other provisions of this contract, the Contractor shall always be responsible for the supervision of his/her employees in the performance of the services required hereunder.

i. During all time on the BEP premises, the Contractor's personnel shall comply with the rules and regulations governing conduct of personnel and operation of the facility.

ii. If the Contractor plans to employ any non-English speaking personnel, he/she shall provide an on-site bilingual supervisor to serve as an interpreter.

iii. H-18 Posting at Work Site Upon commencement of work under this contract, the Contractor shall post a conspicuously located sign at the entrance of each work site. The following information is required to be posted

1. Name of Prime Contractor
2. Name of Prime Contractor's On-Site Supervisor
3. Name of COR
4. Title of Work

iv. **Hazardous Material:** The Contractor is responsible for coordinating and conducting all applicable lead-based paint (LBP) surveys and analyses for the work area(s) prior to demolition. The contractor shall forward all sample analysis results to the BEP LBP Program manager for review at least two weeks prior to the commencement of work. Any LBP abatement required to complete the project shall be conducted/coordinated by the Contractor. In addition, the Contractor shall submit a LBP abatement plan to the BEP LBP Program manager for review and approval two weeks after the Notice to Proceed is given for the commencement of work. The Notice to Proceed ONLY allows the contractor to begin coordinating logistics (i.e., badging, security clearances, etc.). It does not indicate that the abatement can commence. The LBP abatement plan MUST be approved, in writing, by the BEP LBP Program manager prior to the start of the abatement work. The plan MUST be approved, in writing, by the BEP LBP Program manager prior to the start of work.

b. Acceptance of Work

i. Substantial Completion Substantial completion of the work is defined herein as the point at which the work is complete in all respects except for a few minor items which are to be listed on the Contractor's punch list. Substantial Completion is inclusive of all accepted systems being fully operational. The Contractor shall demonstrate that all newly installed systems are fully operational. With the issuance of a substantial completion certificate, executed by the Contracting Officer (CO) and the Contractor, the BEP will take possession of the work and operate the installed systems/equipment. Prior to the issuance of an executed substantial completion certificate the Contractor shall have had all tests completed, witnessed, and approved by the BEP's authorized representative. The provision of all required training,

operating and maintenance manuals, and other such requirements in this SOW shall be a prerequisite to the Contractor's notification requesting a substantial completion walk-through.

ii. The Contractor shall notify the CO and/or COR in writing at least ten (10) calendar days prior to the estimated date of substantial completion and request a substantial completion walk-thru. The letter shall include a dated punch list as developed by the Contractor. The purpose of the walk-through is to review the Contractor's list for accuracy and to identify any additional items needing completion prior to final acceptance. The Contractor, shall, upon receipt of a partial or substantial completion punch list, correct his or her work as required within seven (7) calendar days. The substantial completion walk-through shall be performed by the BEP COR and/or CC.

iii. The date of the BEP's acknowledgement of substantial completion shall establish the Contractor's one year guarantee of workmanship and the manufacturer's product warranties. In the instance of a Partial Substantial Completion Certificate, the previous sentence shall apply to the work identified in the partial substantial completion certificate only. The Contractor shall ensure all warranty requirements, including manufacturers' warranties, in an electronic spreadsheet format that will be provided to the CO and COR.

iv. Prior to requesting Substantial Completion. The Contractor is required to submit

a. Warranties
b. O&M Manuals
c. Records of Waste Disposal (if applicable)
d. Appropriate Testing and Inspection Results
e. Final Punch List with Request for Certification that the Project is Substantially Complete
f. As-Built Drawings

c. **Final Completion:** The Contractor shall inform the CO and/or COR in writing at least seven (7) calendar days prior to the estimated date of the completion of the work and request a final acceptance inspection.

d. **Deliverables:** The Contractor tasks requirements summarize the performance objectives that relate directly to mission essential items. The performance threshold briefly describes the minimum acceptable levels of work required for each requirement. These thresholds are critical to mission success.

Deliverables Table
Submit accurate reports and Deliverables. Division 01 Specification (section 017823 Part 1 no. 2)
Address non-conforming services within 72 hours of notification. Division 01 Specification (section 013200)
Functional testing and commissioning shall consist of operating all associated equipment to ensure the equipment operates correctly. All discrepancies shall be reported to the COR via email within eight working hours of discovery. Division 01 Specification (section 017823 Part 1.8 D.4)

On schedule Project completion Division 01 Specification (section 013200 Part 2 no. 2)

e. Special Delivery Instructions: See Section III where applicable.

f. Performance/ Delivery Period:

The period of performance for this contract shall not exceed more than **730 calendar days** from the issuance of the Notice to Proceed (NTP) to completion.

g. Place/ Location of Performance/ Delivery:

i. Hours of Operation: The contractor is responsible for conducting business, between the hours of, Monday-Friday, between 0800 am and 1700 pm (EST) Monday thru Friday except Federal holidays or when the Government facility is closed due to local or national emergencies, administrative closings, or similar Government directed facility closings. Weekend work is not required but may be necessary depending on the areas in question and approved from the agency. Demo can take place during normal hours. Plan for the possibility of a maximum of 1 weekend if needed. Demo is permitted during normal business hours. The contractor shall coordinate with the government regarding hour of operation for BEP Year End Shut Down, which are generally the days between Christmas and New Years Day. For other than firm fixed price contracts, the contractor will not be reimbursed when the government facility is closed for the above reasons. The Contractor must at all times maintain an adequate workforce for the uninterrupted performance of all tasks defined within this SOW when the Government facility is not closed for the above reasons. When hiring personnel, the Contractor shall keep in mind that the stability and continuity of the workforce are essential.

Federal Holidays

New Years Day	1st day of January
Martin Luther King Jr.'s Birthday	3rd Monday of January
Presidents Day	3rd Monday of February
Memorial Day	Last Monday of May
Juneteenth	19th day of June
Independence Day	4th day of July
Labor Day	1st Monday of September
Columbus Day	2nd Monday of October
Veterans Day	11th day of November
Thanksgiving Day	4th Thursday of November
Christmas Day	25th day of December

ii. Place of Performance: The work to be performed under this contract will be performed at BEP's DCF 14th & C Streets, S.W., Washington, D.C.

h. Special Requirements.

i. Security Requirements: Contractor personnel (to include subcontractors) performing work under this contract shall obtain a permanent BEP contractor badge for unescorted access. Until contractor personnel can obtain permanent badges, they shall be escorted by approved BEP personnel. Contractors that do not have badges and need access into the building at any time

will be required to obtain a Visitor's Badge at either the 13th Street or 14th Street entrance and be escorted by Contractors Badged Personnel. They will need a government issued Photo I.D. and present it to the Police Officer at the desk after clearing the X-Ray Belt Machine and Magnetometers. Access to BEP computers and networks shall not be granted to personnel who have not been cleared for access by the BEP Office of Security and approved for computer network access. If computer access is required, the BEP escort shall contact the BEP Service Desk to request support from the CIO Directorate.

ii. Physical Security The Contractor shall be responsible for safeguarding all government equipment, information, and property provided for Contractor use under this ordering agreement. This ordering agreement requires access into the Controlled Industrial Area of the BEP. The Contractor personnel shall recognize that BEP deals with sensitive information and other proprietary information, which are valuable, special, and unique assets. All information pertinent to BEP's information technology resources is highly sensitive and may not be released to third parties without the express, written consent of BEP. This information requires strict control procedures to preclude unauthorized disclosure or dissemination. Information pertinent to personnel is protected from unauthorized disclosure under the provisions of the Privacy Act of 1974 (5 U.S.C. 552a). Contractor personnel shall not divulge, disclose, or communicate in any manner, any information to any third party without prior written consent of the BEP. Contractor personnel shall protect information and treat it as strictly confidential and will abide by the security requirements of BEP. A violation of this paragraph shall be a material violation of this ordering agreement. Should the Contractor receive or suspect they have learned of sensitive or classified information, the Contractor shall identify and report suspicions to the COR. Contractor personnel performing work under this ordering agreement shall be subjects of background investigations, conducted by BEP's Office of Personnel Security, prior to the start of work under this ordering agreement. Each Contractor employee performing under this ordering agreement must successfully complete the Government-required background check form at time of the proposal submission and must maintain the level of security required for the life of the ordering agreement.

Key Control: The Contractor shall establish and implement methods of making sure all keys/key cards issued to the Contractor by the Government are not lost or misplaced and are not used by unauthorized persons. NOTE All references to keys include key cards. No keys issued to the Contractor by the Government shall be duplicated. The Contractor shall develop procedures covering key control that shall be included in the Quality Control Plan. Such procedures shall include the turn-in of any issued keys by personnel who no longer require access to locked areas. The Contractor shall immediately report any occurrences of lost or duplicate keys/key cards to the Contracting Officer.

In the event keys, other than master keys, are lost or duplicated, the Contractor shall, upon the direction of the Contracting Officer, re-key or replace the affected lock or locks; however, the Government, at its option, may replace the affected lock or locks or perform re-keying. When the replacement of locks or re-keying is performed by the Government, the total cost of re-keying or the replacement of the lock or locks shall be deducted from the monthly payment due the Contractor. In the event a master key is lost or duplicated, all locks and keys for that system shall be replaced by the Government and the total cost deducted from the monthly payment due to the Contractor.

The Contractor shall prohibit the use of government-issued keys/key cards by any persons other than the Contractor's employees. The Contractor shall prohibit the opening of locked areas by Contractor employees to permit entrance of persons other than Contractor employees engaged in the performance of assigned work in those areas, or personnel authorized entrance by the Contracting Officer.

iii. **Special Qualifications**: The Contractor and all Sub-Contractors shall have extensive experience with installing HVAC systems for industrial and federal buildings. The experience shall specialize in mechanical contracting with an emphasis on large HVAC installations for existing buildings. Contractor or sub-contractor performing integration to SCADA shall be certified as Citect SCADA Certified Expert (CSCE). Contractor or sub-contractor performing controls installation shall be an FX ASI (Facility Explorer Authorized System Installer.) The contractor shall be responsible for ensuring all employees possess all required licenses and training for operating equipment and vehicles used in the execution of this contract. Use of any chemicals shall meet EPA guidelines, federal regulations, and District of Columbia (DC) Department of Health requirements.

iv. **Post-Award Conference/Periodic Progress Meetings**: The Contractor agrees to attend any post-award conference convened by the contracting activity or contract administration office in accordance with Federal Acquisition Regulation Subpart 42.5, Post Award Orientation. The Contracting Officer, COR, and other Government personnel, as appropriate, may meet periodically with the contractor to review the contractor's performance. At these meetings the Contracting Officer will apprise the contractor of how the government views the contractor's performance and the contractor will apprise the Government of problems, if any, being experienced. Appropriate action shall be taken to resolve outstanding issues. These meetings shall be at no additional cost to the Government.

v. **Contracting Officer Representative (COR)**: The COR will be identified by a separate letter by the CO. The COR monitors all technical aspects of the contract and assists in contract administration. The COR is authorized to perform the following functions assure that the Contractor performs the technical requirements of the contract; perform inspections necessary in connection with contract performance; maintain written and oral communications with the Contractor concerning technical aspects of the contract; issue written interpretations of technical requirements, including Government drawings, designs, and specifications; monitor Contractor's performance and notify both the Contracting Officer and Contractor of any deficiencies; coordinate availability of government furnished property; and provide site entry of Contractor personnel. A letter of designation issued to the COR, a copy of which is sent to the Contractor, states the responsibilities and limitations of the COR, especially with regard to changes in cost or price, estimates, or changes in delivery dates. The COR is not authorized to change any of the terms and conditions of the resulting order.

vi. **Key Personnel:** The following personnel are considered key personnel by the government specialized technical personnel, project manager and superintendent. The contractor shall provide a contract manager who shall be responsible for the performance of the work. The name of this person and an alternate who shall act for the contractor when the manager is absent shall be designated in writing to the contracting officer. The contract manager or alternate shall have full authority to act for the contractor on all contract matters relating to daily operation of this contract. The contract manager or alternate shall be available between 0700 am to 1600 pm, Monday thru Friday except Federal holidays or when the government facility is closed for administrative reasons.

Resumes of Key Personnel shall show demonstrated experience in the position they're being considered for. The continued full-time availability of Key Personnel shall be formally confirmed in writing to the Government at time of award. The following personnel are considered "Key" Personnel.

vii. **Specialized Technical Personnel:** Specialized Technical Personnel shall have a bachelor's degree from a U.S. accredited university or college with a major in engineering, architecture, or related technical field. Personnel shall demonstrate work experience, licenses (Professional Engineering) and technical competence of electrical, mechanical, structural/civil, architectural, fire protection, or environmental requirements covering the last five (5) years.

Project Manager: The Project Manager shall have a bachelor's degree from a U.S. accredited university or college with a major in engineering, architecture, or related technical field, and/or at least five years' experience in managing architectural, engineering or construction projects of similar size and scope and five (5) years' experience in Historical Government Buildings. The Project Manager shall be employed by the Prime Contractor.

Project Superintendent: The Project Superintendent shall have a minimum of five (5) years' experience as a superintendent on projects similar in size and scope, and five (5) years' experience in Historical Government Buildings. be employed by the Prime Contractor. The project superintendent shall have overall responsibility for all operations on the jobsite.

Key Personnel Substitution Process: The Government reserves the right to request replacement of any personnel who are deemed incompetent, careless, unsuitable or otherwise objectionable, or whose continued use under this contract is deemed contrary to the best interests of the Government. The COR will give notice, in writing, to the Contractor POC of such need for substitution. Within 48 hours of receipt of notification, the Contractor POC shall provide personnel replacement resumes to the COR. The COR will notify the Contractor POC, in writing, if the Contractor provided replacement personnel is approved. The COR will notify the Contracting Officer (CO) of the approval, and CO will issue a modification to formally replace the personnel.

During the performance of this contract, the contractor shall make no substitutions of key personnel unless the substitution is necessitated by illness, death, or termination of employment. The contractor shall notify the COR of this contract no later than five (5) calendar days after the occurrence of any of these events. The contractor shall include the circumstances necessitating the proposed replacement of the key personnel and shall provide to the Contracting Officer the name, resume of the proposed replacement. All replacement personnel require written approval from the Contracting Officer prior to their beginning work under this contract. When required, the contractor shall provide replacement employees that comply with Sections 3.27.8 through 3.27.8.3 of this SOW. Any disputes regarding the substitution of key personnel after resume review will be decided by the Contracting Officer. The Government reserves the right to require the contractor to reassign key contractor employees who are deemed incompetent, careless, unsuitable, or otherwise objectionable, or whose continued use under this contract is deemed contrary to the best interests of the Government. The Contracting Officer will provide a written notice to the contractor of such reassignment.

viii. **Identification of Contractor Employees:** All contract personnel (to include subcontractors) attending meetings, answering Government telephones, and working in other situations where their contractor status is not obvious to third parties are required to identify themselves as such to avoid creating an impression in the minds of members of the public that they are Government officials. They must also ensure that all documents or reports produced by Contractors are suitably marked as Contractor products or that Contractor participation is appropriately disclosed. All Contractor personnel shall clearly identify themselves as a Contractor and shall be required to wear identification (ID) badges in a visible location on the front of exterior clothing, above the waist, (except when safety or health reasons prevent such placement). ID badges shall be worn at all times when performing work. ID badges shall have an identification picture and shall state the name of the individual and the company they represent. A listing of issued identification cards shall be furnished to the Contracting Officer prior to the contract performance date and updated as needed to reflect Contractor and Subcontractor personnel changes. Contractor personnel are prohibited from possessing weapons, firearms or ammunition, on themselves or within their Contractor-owned or privately owned vehicle while on BEP premises.

ix. **Supervision of Contractor Employees** The Government will not exercise any supervision or control over Contractor or subcontractor employees while performing work under the contract. Such employees shall be accountable solely to the Contractor, not the Government. The Contractor, in turn, shall be accountable to the Government for Contractor or subcontractor employees.

x. **Data Rights:** The Government has unlimited rights to all documents/material produced under this contract. All documents and materials, to include the source codes of any software, produced under this contract shall be Government owned and are the property of the Government with all rights and privileges of ownership/copyright belonging exclusively to the Government. These documents and materials may not be used or sold by the Contractor without written permission from the Contracting Officer. All materials supplied to the Government shall be the sole property of the Government and may not be used for any other purpose. This right does not abrogate any other Government rights.

xi. **Organizational Conflict of Interest**: Contractor and subcontractor personnel performing work under this contract may receive, have access to or participate in the development of proprietary or source selection information (e.g., cost or pricing information, budget information or analyses, specifications or work statements, etc.) or perform evaluation services which may create a current or subsequent Organizational Conflict of Interests (OCI) as defined in FAR Subpart 9.5. The Contractor shall notify the Contracting Officer immediately whenever it becomes aware that such access or participation may result in any actual or potential OCI and shall promptly submit a plan to the Contracting Officer to avoid or mitigate any such OCI. The Contractor's mitigation plan will be determined to be acceptable solely at the discretion of the Contracting Officer and in the event the Contracting Officer unilaterally determines that any such OCI cannot be satisfactorily avoided or mitigated, the Contracting Officer may affect other remedies as he or she deems necessary, including prohibiting the Contractor from participation in subsequent contracted requirements which may be affected by the OCI.

xii. **Non-Personal Service statement:** The Contractor employees performing services under this Contract will be controlled, directed, and supervised at all times by management personnel of the Contractor. Contractor management will ensure that employees properly comply with the performance work standards outlined in Contract. Contractor employees will perform their

duties independent of, and without the supervision of, any Government official or other Government Contractor. The tasks, duties, and responsibilities set forth in this Contract may not be interpreted or implemented in any manner that results in any Contractor employee creating or modifying Federal policy, obligating the appropriated funds of the United States Government, overseeing the work of Federal employees, providing direct personal services to any Federal employee, or otherwise violating the prohibitions set forth in Parts 7.5 and 37.1 of the FAR.

SECTION IV: Applicable Documents

Attachment 1 75D-04.0-08 Environmental, Health and Safety (EHS) Form Construction Safety Permit (after award)
Attachment 2 75D-07.0-04 EMS Document EHS Contractor Requirements (mandatory compliance, after award)
Attachment 3 75D-04.0-11 Fall Protection Procedure (mandatory compliance, after award)
Attachment 4 75D-04.0-01 Emergency Action and Fire Prevention Plan (mandatory compliance, after award)
Attachment 5 Mutual Multi Party Non-Disclosure Agreement(s) (Upon request by the government)
Attachment 6 Drawings A (After award)
Attachment 7 Drawings B: M-DCA-09-1001023-HV (After award)

SECTION V: Requirements for Handling Sensitive, Classified, and/or Proprietary Information

Complete appropriate NDA's per DTAP policy as required.

SECTION VI: Government Furnished Resources and Information

GOVERNMENT FURNISHED ITEMS AND SERVICES: All property equipment and services shall be provided by the contractor. If the contractor requires access for dropping off or picking up materials for specific jobs, entrance to the loading dock can be granted with advanced notice to the COR. No parking shall be provided by BEP for other purposes.

CONTRACTOR FURNISHED ITEMS AND RESPONSABILITIES: The contractor shall supply all equipment required to perform the specific tasks listed in Section III. Any waste created by the contractor shall be disposed of according to local and federal regulations. Supplies used shall be of the type, composition, quality, and rate standard in the professional industry for industrial hygiene services. All supplies or chemicals brought onsite shall be reported to and approved by the COR or his representative prior to use. Labels and Safety data sheet (SDS) sheets shall also be provided. X-ray diffraction (XRF) and other industrial hygiene equipment shall be calibrated and maintained according to federal requirements and manufacturer recommendation.

(END OF DOCUMENT)